Figure 4.1. Integrated Concept Map: Cellular Neuroscience: Neurons, Glia, Myelin, and Neural Tissue Responses

Chapter topics and a reading framework

- 4.1 Cell theory and nervous tissue
- 4.2 Neuronal structure
- 4.3 Membrane potentials
- 4.4 Action potentials
- 4.5 Astrocyte functions
- 4.6 Microglial surveillance
- 4.7 Oligodendrocytes and myelin
- 4.8 Blood-brain barrier
- 4.9 Cellular stress responses
- 4.10 Cellular repair and adaptation

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 4, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.